TECHNICAL WORKFLOW GUIDE

Employee Group Insurance (EGI)

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1. System Overview & Core Constants

This section defines the immutable technical constants that govern all financial calculations, state transitions, and commission logic throughout the EGI Portal. Every workflow in this document operates within these boundaries.

1.1 Technology Stack

Component	Details
Frontend	Angular 17+ — Standalone Components, Signals for state management, Lazy-loaded routes
Backend	.NET 8 Web API — Clean Architecture, Repository + Unit of Work patterns
Database	Microsoft SQL Server — Entity Framework Core (Code-First migrations)
PDF Engine	QuestPDF — Invoice PDFs and Claim Settlement Documents
Auth	JWT (JSON Web Tokens) with BCrypt password hashing — role-bound claims
Background	InsuranceAutomationWorker (IHostedService) — runs every 24 hours

1.2 Financial & Timeline Constants

! These constants are sourced directly from BusinessRules.cs and must not be overridden by business logic elsewhere.

Constant	Value / Rule
Grace Period	7 days from Invoice Date to Due Date — no penalty within this window
Soft Suspension	Day 10 (3 days overdue) — Penalty warnings triggered; customer notified
Hard Suspension	Day 14 total (7 days overdue) — Policy Inactivated; Payments Blocked; Claims Blocked
Penalty Rate	10% flat applied to total invoice amount on the Due Date (Day 7)
Annual Discount	5% discount on total calculated premium for Annually-billed policies
Min Partial Pay	MAX(50% of Total Invoice, Remaining Balance) — payments below this are rejected

Password Length	10 high-entropy characters for auto-generated temporary credentials

1.3 Commission Tiers (BusinessRules.cs)

> **i** Rates are INVERTED by design — smaller businesses receive higher commission rates to incentivize agent effort on harder-to-close accounts.

Tier	Insured Lives (Members + Active Dependents)	Commission Rate
Small Business	≤ 50 lives	8%
Medium Business	51 – 250 lives	6%
Large Business	251 – 1,000 lives	4%
Enterprise	> 1,000 lives	2%

> **!** Commission is triggered ONLY on successful payment receipt — not on policy creation or endorsement approval. Every calculation creates a CommissionCalculation AuditLog entry.

1.4 Premium Multipliers (PolicyEndorsementService.cs)

Applied during endorsement adjustment calculations to determine the Annual Rate Impact per insured person:

Insured Person Type	Multiplier
Primary Member (Employee)	1.0× Base Premium
Spouse	0.8× Base Premium
Child	0.4× Base Premium
Parent (Father / Mother)	1.2× Base Premium

2. Workflow 1 — Customer Registration & Onboarding

Goal: Onboard a new corporate customer through either self-service or agent-assisted registration. Both paths converge at Admin KYB approval and agent assignment.

Actors	Customer, Agent, Admin, System
Entry Points	Public signup page (Path A) or Agent dashboard (Path B)
Final State	Customer approved, assigned to Agent, policy-ready
Key Guard	KYB documents must be Admin-approved before any policy can be issued

2A. Path A — Customer Self-Registration

REGISTERED → **DRAFT** → **PENDING** → **APPROVED** → **ASSIGNED**

1	Customer navigates to the public signup page and registers with their email and a chosen password. Account is created in DRAFT status.
2	Customer completes company profile: Name, Address, Phone, Business Category, and Company Registration Number.
3	**KYB Upload:** Customer uploads mandatory verification documents (Company Registration Certificate, Tax Documents, Business License). Status transitions to **PENDING**.
4	**Admin Review:** Admin manually inspects uploaded KYB documents in the verification queue.
5	**Strike Rule:** Admin can REJECT the submission. The customer may re-submit corrected documents. After **3 rejections**, the account is **PERMANENTLY BLOCKED** — no further appeal is possible. RejectionCount is stored on CorporateClient.
6	**On Approval:** Status transitions to APPROVED. If no Agent is yet assigned, the **Least Loaded Agent algorithm** runs automatically — assigns the customer to the Agent with the lowest current active customer count.
7	Customer receives an approval notification. Account is now fully active and policy-ready.

2B. Path B — Agent-Created Customer

AGENT CREATES → PENDING → APPROVED → CREDS SENT → FORCE RESET

1	Agent creates a Customer account directly from the Agent Dashboard — enters company details on behalf of the client.
2	**Agent uploads KYB documents** on the customer's behalf. Customer is **permanently linked** to this Agent from creation — the Least Loaded algorithm is bypassed.
3	Account enters Admin's verification queue in PENDING status. Admin reviews the agent-uploaded documents.
4	**On Rejection:** The creating Agent is notified to re-upload corrected documents. The same 3-strike limit applies.
5	**On Approval:** System generates a secure 10-character temporary password using high-entropy characters and emails it to the Customer's registered email address.
6	**Forced Password Reset:** Customer logs in with temporary credentials and is **strictly redirected** to a password change screen. The **requirePasswordChange** flag is set to true. No dashboard access is granted until this step is completed.
7	Customer gains full dashboard access. Account is policy-ready and permanently associated with the creating Agent.

i Key Difference: In Path A, agent assignment is dynamic (Least Loaded algorithm). In Path B, the agent-customer bond is permanent from account creation.

3. Workflow 2 — Internal Staff Provisioning

Goal: Securely add Agents and Claims Officers to the system with enforced first-login credential hygiene.

Actors	Admin, System, New Staff Member (Agent or Claims Officer)
Trigger	Admin initiates from Admin Dashboard → Staff Management
Output	Active staff account with role claim + forced password reset on first login
Roles Created	Agent, ClaimsOfficer

ADMIN INITIATES → **ACCOUNT CREATED** → **EMAIL SENT** → **FIRST LOGIN** → **ACTIVE**

1	**Admin Input:** Admin enters the new staff member's full Name, Email address, and selects Role (**Agent** or **ClaimsOfficer**) in the Admin Dashboard.
2	**Credential Generation:** System generates a secure **10-character temporary password** using high-entropy characters (uppercase, lowercase, digits, special characters).
3	**Account Creation:** User record is created with the specified Role JWT claim and **MustChangePassword = true**. Account status is Active from creation.
4	**Email Dispatch:** System sends an automated email to the new staff member containing their login email and temporary password.
5	**First Login Guard:** On initial login, the system detects **MustChangePassword = true** and strictly redirects the staff member to the password update screen. No application features are accessible until the password is changed.
6	Staff member sets a new personal password. MustChangePassword flag is cleared. Full role-appropriate dashboard access is granted.

! Security Note: Temporary passwords are single-use. The MustChangePassword flag cannot be bypassed via any API endpoint — it is enforced server-side at the middleware level.

4. Workflow 3 — Member & Dependent Management

Goal: Manage the insured headcount, coverage limits, and dependent relationships that drive premium calculations and commission tier assignments.

Actors	Customer (HR Manager), System
Trigger	Policy assignment approval, or mid-term Endorsement (Add/Remove)
Output	Updated Lives count, refreshed CommissionTier, correct Sum Insured allocation
Storage	Soft deletes only — Members/Dependents use IsActive = false to preserve history

4.1 Lives Count Definition

> **i** Lives = Count of Active Members + Count of Active Dependents. This total drives the CommissionTier assigned to the Agent for ALL future commission calculations on this policy.

4.2 Sum Insured Allocation

Insured Type	Sum Insured Rule
Member Sum Insured	100% of the Plan's Health Coverage limit — full allocation.
Dependent Sum Insured	Calculated based on Relationship Type using the multiplier table.
Shared Pool	Dependent claims draw from the primary Member's unified family Sum Insured pool — preventing total family liability from exceeding the policy cap.
Claim Deduction	Approved claim amounts are permanently deducted from the claimant's remaining Sum Insured for the current policy year.

4.3 Premium Multipliers (Per Dependent Type)

Relationship Type	Multiplier	Note
Primary Member (Employee)	1.0×	Full Base Premium — baseline for all calculations
Spouse	0.8×	80% of Base Premium
Child	0.4×	40% of Base Premium — lowest risk weighting
Parent (Father / Mother)	1.2×	120% of Base Premium — highest risk weighting

4.4 Inactivation Cascade Rule

!	Removing (inactivating) a Member recursively inactivates ALL their linked Dependents. This is a hard cascade — not a soft suggestion. IsActive = false is set on the Member and all dependent records simultaneously.

1	Customer submits a RemoveMember endorsement (or Admin/Agent forces inactivation).
2	System sets Member.IsActive = false on the primary member record.
3	System queries all Dependent records where MemberId = this member. Sets Dependent.IsActive = false on each.
4	All active Claims linked to this Member or their Dependents are flagged — no new claims can be submitted.
5	Lives count is decremented. System re-evaluates CommissionTier based on new total Lives count.
6	A prorated Financial Credit is calculated for unused coverage days and queued in the Waterfall Credit system.

5. Workflow 4 — Invoicing & Payment (The Waterfall Logic)

Goal: Track the complete financial lifecycle from automated invoice generation through penalty enforcement, policy suspension, and payment settlement including commission calculation.

Actors	System (Automated Worker), Customer, Agent
Trigger	Policy start date, BillingFrequency cycle, Endorsement approval
Invoice ID	Format: INV-[Year]-[GUID] for standard invoices; INV-[Year]-[GUID]-ADJ for adjustment invoices
Billing Modes	Monthly (no discount) | Annually (5% discount applied at generation)

5.1 Invoice Lifecycle & Timeline

GENERATED → **PENDING** → **OVERDUE** → **SUSPENDED** → **PAID / PARTIAL**

Timeline	Event	System Action
Day 0	Invoice generated by background worker.	Status: PENDING. Format: INV-[Year]-[GUID]
Day 1–7	Grace period — customer can pay with no penalty.	Status: PENDING. No action required.
Day 7	Due Date reached. Invoice still unpaid.	**10% penalty applied. Status → OVERDUE.**
Day 10	3 days overdue. Soft suspension warning.	Daily countdown notifications sent to Customer.
Day 14	7 days overdue. Hard suspension threshold reached.	**Policy → INACTIVE. Payments BLOCKED. Claims BLOCKED.**

! Once a Policy reaches INACTIVE status (Day 14), that specific invoice's payment is also BLOCKED. The only resolution path is Admin/Agent manual intervention.

5.2 Partial Payment Rules

i Partial payments are natively supported. The system does NOT require full payment in one transaction. However, a minimum threshold is strictly enforced.

Rule	Detail
Minimum Payment Rule	MAX(50% of Total Invoice, Remaining Balance) — payments below this are rejected outright.

Status on Partial Pay	Invoice status moves to PARTIALLY PAID if payment < 100% of total.
Status on Full Pay	Invoice status moves to PAID. Policy remains/becomes Active.
Multiple Payments	Multiple partial payments are allowed until the invoice is fully settled.

5.3 Commission Calculation on Payment

1	**Trigger:** A payment event fires (Full OR Partial). Commission is NEVER calculated at policy creation.
2	**Tier Lookup:** System reads the current **CommissionTier** for the Agent associated with this policy's corporate client.
3	**Math:** Earned = PaymentAmount × CurrentCommissionTier. Example: ₹10,000 payment on a Small Business policy → ₹10,000 × 8% = **₹800 commission.**
4	**Audit Entry:** Every commission calculation creates an immutable **CommissionCalculation** AuditLog entry stamping: AgentId, PolicyId, PaymentAmount, Rate, EarnedAmount, Timestamp.
5	**Re-Categorization:** If an Endorsement changed the Lives count and moved the policy to a different tier, the **new rate applies to all future payments**. Past calculations are not retroactively changed.

5.4 The Waterfall Credit System

i	When a RemoveMember or RemoveDependent endorsement results in a negative adjustment (a refund owed to the customer), the system does not issue a direct cash refund. Instead, it applies a credit against future invoices.

1	Endorsement approved with negative premium adjustment (removal of covered person mid-period).
2	System calculates the prorated Financial Credit: BasePremium × Multiplier × (DaysRemaining / 365).
3	**Waterfall Logic:** System queries all PENDING and OVERDUE invoices for this policy, ordered chronologically (oldest first).
4	Credit is applied against the soonest available invoice first. If credit exceeds one invoice's balance, the remainder cascades to the next invoice in chronological order.
5	If credit exceeds all outstanding invoices, the remaining balance is held as a credit on the account for future invoices.

6. Workflow 5 — Mid-Term Endorsements (Policy Revisions)

Goal: Handle workforce changes (hires, exits, dependent additions/removals) mid-contract with mathematically accurate prorated premium adjustments.

Actors	Customer, Agent, Admin, System
Trigger	HR event — new hire, termination, marriage, birth of child, etc.
Endorsement Types	AddMember, RemoveMember, AddDependent, RemoveDependent
Output	Adjustment Invoice (positive) or Waterfall Credit (negative) + updated Lives count

SUBMITTED → **PENDING** → **UNDER REVIEW** → **APPROVED** → **REJECTED**

6.1 Submission & Review

1	**Customer Submits:** Customer selects endorsement type (AddMember, RemoveMember, AddDependent, RemoveDependent) from their policy dashboard.
2	Customer provides endorsement details: member/dependent name, date of birth, relationship, effective date, and justification. Data is serialized and stored as JSON on the PolicyEndorsement record.
3	Endorsement status set to PENDING. Assigned Agent is notified (notification + email).
4	**Agent Review:** Agent inspects the endorsement JSON data and either Approves or Rejects.
5	**On Rejection:** Agent provides a mandatory Rejection Reason. Customer is notified and may resubmit a corrected request.

6.2 Prorated Adjustment Calculation (The Exact Day Rule)

i	All calculations use exact calendar days — not rounded months. This ensures mathematical fairness and matches the PolicyEndorsementService.cs implementation.

Step 1 — Calculate Annual Rate Impact

AnnualRateImpact = BasePremium × DependentMultiplier Example: BasePremium = ₹12,000 | Adding a Spouse → ₹12,000 × 0.8 = **₹9,600 Annual Rate Impact**

Step 2 — Apply Billing Frequency

Billing Mode	Calculation Scope
Monthly Payers	Adjustment charged only for the remaining days of the CURRENT billing month. Future months are handled by standard invoice updates (new member included in next regular invoice).
Annual Payers	Adjustment charged for the remaining days of the full POLICY YEAR.

Step 3 — Final Adjustment Math

i	Adjustment = AnnualRateImpact × (DaysRemaining / 365) Example: Adding a Spouse on Day 183 of a 365-day annual policy. → ₹9,600 × (182 / 365) = ₹9,600 × 0.499 = ₹4,788 adjustment invoice generated.

6.3 Financial Output on Approval

Adjustment Direction	System Action
Positive Adjustment (Add Member/Dependent)	System generates an adjustment invoice: INV-[Year]-[GUID]-ADJ. Customer must pay this additional premium for the remaining coverage period.
Negative Adjustment (Remove Member/Dependent)	System computes the credit for unused coverage days and triggers the Waterfall Credit system — applied against the soonest PENDING/OVERDUE invoice.

6.4 Re-Categorization Trigger

1	Endorsement is approved by Agent/Admin.
2	System re-counts total Lives = Active Members + Active Dependents for this policy.
3	System evaluates the new Lives count against the CommissionTier thresholds.
4	**If tier changes:** BusinessCategory is updated (e.g., Small → Medium). CommissionTier is updated for ALL future payment calculations on this policy.
5	Past commission calculations are NOT retroactively adjusted — only future payments use the new rate.

7. Workflow 6 — Claim Adjudication & Settlement

Goal: Process medical payout requests against individual and family coverage limits with strict concurrency control to prevent double-processing.

Actors	Customer, Claims Officer, System
Trigger	Customer submits a claim with supporting documents and requested amount
Concurrency	Exclusive lock (Take/Release) prevents multiple officers processing the same claim
Output	Approved claim with QuestPDF Settlement Document OR Rejected with mandatory reason

SUBMITTED → **PENDING** → **IN REVIEW** → **APPROVED** → **REJECTED**

7.1 Hard Gates — Pre-Submission Validation

! All three gates are checked BEFORE a claim can be submitted. Any single failure blocks the claim entirely.

Gate	Check	Block Condition
Gate 1	Policy Status	Claim is BLOCKED if Policy is INACTIVE (overdue > 14 days). Customer must resolve outstanding invoices first.
Gate 2	Member/Dep. Status	Claim is BLOCKED if the claimant (Member or Dependent) has IsActive = false.
Gate 3	ClaimType Coverage	Claim is BLOCKED if the requested ClaimType (Health, Life, Accident, CriticalIllness) is not included in the assigned Insurance Plan's PlanCoverage records.

7.2 Coverage Enforcement

1	**CoveredGroup Check:** If the claimant is a Dependent (e.g., Child), the system verifies that the plan's CoveredGroup includes 'Family'. A Child cannot claim under an 'Employee Only' plan.
2	**Cumulative Limit Check:** System calculates: PreviousApprovedAmount + CurrentClaimAmount. This total must be ≤ the CoverageLimit for the specific ClaimType on the plan.
3	If either check fails, the claim is rejected at submission with a system-generated error — it never enters the adjudication queue.

7.3 Concurrency Control — Take / Release Mechanism

Action	Description
TAKE (Lock)	Claims Officer selects 'Take' on a PENDING claim. Status moves to IN REVIEW. A lock record is created binding this claim to this officer. All other officers see the claim as locked and cannot Take it.
RELEASE	If the officer cannot complete adjudication, they select 'Release'. The lock is removed. Status reverts to PENDING. Claim re-enters the open queue for any officer to Take.
Timeout (Future)	Planned: Locks automatically expire after a configurable period if neither Approved nor Released.

7.4 Approval Impact

1	**Sum Insured Deduction:** The sanctioned amount is **permanently deducted** from the claimant's remaining SumInsured for the current policy year. This cannot be reversed.
2	**For Dependents:** The deduction is applied against the primary Member's unified family SumInsured pool — ensuring total family claims never exceed the policy cap.
3	**QuestPDF Generation:** System dynamically generates a PDF Settlement Document including: Claim ID, Member details, Sanctioned Amount, Remaining Coverage, and Officer signature block.
4	**Email Notification:** Settlement PDF is attached to an automated email notification sent to the Member's registered email address.
5	**Audit Log:** An immutable AuditLog entry is created: ClaimsOfficerId, ClaimId, SanctionedAmount, Timestamp, NewValues.

! Rejection Rule: Any transition to REJECTED status requires a mandatory RejectionReason string. The API enforces this — a rejection without a reason string returns a 400 Bad Request.

8. Workflow 7 — Background Automation Service

Goal: Automated daily system maintenance — handles invoice generation, overdue enforcement, penalty application, policy suspension, and expiry without manual intervention.

Service Name	InsuranceAutomationWorker (implements IHostedService)
Frequency	Every 24 hours — runs as a .NET background hosted service
Tasks	3 sequential tasks: Generate Due Invoices → Overdue Sweep → Policy Expiration
Trigger	Automatic — no manual intervention required

Task A — Generate Due Invoices (GenerateDueInvoicesAsync)

1	Worker scans all ACTIVE policies where the next billing period starts within 30 days.
2	Reads BillingFrequency (Monthly or Annually) to determine the invoice amount and due date.
3	**Annual Discount:** If BillingFrequency = Annually, system applies a 5% discount to the total calculated premium before generating the invoice.
4	Generates Invoice record: INV-[Year]-[GUID], sets Amount, DueDate, Status = PENDING.
5	Notifies the Customer (in-app notification) that a new invoice has been generated.

Task B — Overdue Sweep (MarkOverdueInvoicesAsync)

1	Worker scans all PENDING invoices and checks elapsed days since DueDate.
2	**Day 7 (Penalty):** Applies 10% flat penalty to the total invoice amount. Invoice status → OVERDUE. Customer notified.
3	**Day 10 (Soft Warning):** Sends daily countdown notification to Customer warning of impending suspension.
4	**Day 14 (Hard Suspension):** Sets Policy.Status = INACTIVE. Blocks payment processing for this invoice. Blocks all claim submissions for this policy. Agent notified.
5	Creates AuditLog entry for each status transition: PolicyId, PreviousStatus, NewStatus, Timestamp.

Task C — Policy Expiration (ExpireOldPoliciesAsync)

1	Worker scans all ACTIVE and INACTIVE policies and checks the EndDate field.
2	Any policy where EndDate <= today is flagged as EXPIRED.
3	Expired policies block all new claim submissions and endorsement requests.
4	Renewal workflow (if applicable) is triggered — out of scope for this version.

> **i** Task execution order matters: Generate → Sweep → Expire. Running Expire before Sweep could cause edge cases where a policy is expired before its final overdue invoice is flagged.

9. Database & Audit Trail Standards

These standards apply universally across all 7 workflows. Every state-changing operation must comply with both the soft delete policy and the audit logging requirement.

9.1 Audit Logging — Every Write Operation

> **!** Every write operation (status changes, calculations, locks, approvals, rejections) MUST create an immutable AuditLog entry. This is a non-negotiable compliance requirement.

Required AuditLog Fields

Field	Description
UserId	ID of the user who performed the action (or 'SYSTEM' for automated worker tasks).
Entity	Name of the affected entity (e.g., Policy, Invoice, Claim, Endorsement, Member).
Action	Description of the action performed (e.g., StatusChanged, PenaltyApplied, ClaimApproved).
Timestamp	UTC timestamp of the action.
NewValues	JSON snapshot of the entity's new state after the change.

Audit-Triggering Operations

Operation	What Changes	Triggered By
Policy Status Change	ACTIVE → INACTIVE / EXPIRED	Worker or Admin action
Invoice Penalty Applied	Amount updated + Status → OVERDUE	Worker task (Day 7)
Commission Calculated	CommissionCalculation record created	Payment event
Claim Status Transition	PENDING → IN REVIEW → APPROVED/REJECTED	Claims Officer action
Endorsement Approved	Roster + financials updated	Agent/Admin action
Claim Lock Acquired	Status → IN REVIEW + OfficerId bound	Claims Officer Take
Claim Lock Released	Status → PENDING + OfficerId cleared	Claims Officer Release
KYB Approved/Rejected	VerificationStatus updated + RejectionCount++	Admin action
Password Reset	PasswordHash updated	User or Admin action
Staff Account Created	User record with Role claim created	Admin action

9.2 Soft Delete Policy

> **i** Members and Dependents NEVER use hard deletes. IsActive = false (soft delete) is the ONLY removal mechanism. This preserves all claim history, coverage records, and audit trails.

Entity	Soft Delete Behaviour
Member Removal	Member.IsActive = false. All linked Dependents cascade to IsActive = false simultaneously.
Dependent Removal	Dependent.IsActive = false. Primary Member record is unaffected.
Claim History	Inactive Members/Dependents retain all historical claim records for compliance and reporting.
Coverage History	Sum Insured deduction history is preserved even after member inactivation.

9.3 Invoice Naming Convention

Invoice Type	Format
Standard Invoice	INV-[Year]-[GUID] — generated by the monthly/annual billing cycle
Adjustment Invoice	INV-[Year]-[GUID]-ADJ — generated by an approved AddMember/AddDependent endorsement
Credit Note	Applied internally via Waterfall Credit system — no separate invoice number